Preventive Maintenance Manual - TX-200 CNC Turning Centre

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1. Purpose

This manual defines the preventive maintenance (PM) activities, operating parameters and common alarms for the TX-200 CNC turning centres (machines CNC-04 to CNC-09) used on the shaft machining line.

2. Safety First

Before starting any maintenance activity other than daily operator checks, the machine must be isolated as per **SAF-SOP-003 Lockout/Tagout**. Never open the electrical cabinet or hydraulic circuit on a running or energised machine.

3. Normal Operating Parameters

Parameter	Normal range	Check method
Hydraulic system pressure	3.5 - 4.0 MPa (35 - 40 bar)	Panel pressure gauge
Chuck clamping pressure	2.0 - 2.5 MPa	Chuck pressure gauge
Coolant concentration	6 - 8 %	Refractometer reading x 1.0 factor
Coolant tank level	Between MIN and MAX marks	Sight glass
Way lube oil	ISO VG 68	Visual, top-up as needed
Spindle runout (at chuck face)	0.005 mm max.	Dial indicator
Compressed air supply	5.5 - 6.5 bar	FRL unit gauge

4. Preventive Maintenance Schedule

Frequency	Activity	Done by
Daily (start of shift)	Check hydraulic pressure, coolant level, way lube level, air pressure; clean chips from slideways	Operator
Weekly	Check coolant concentration; clean coolant tank filter; check chuck jaw wear	Operator
Monthly	Check spindle runout; inspect hydraulic hoses for leaks or cracks; clean electrical cabinet fan filter	Maintenance technician
Quarterly	Replace hydraulic return filter; check turret alignment; check backlash on X and Z axes	Maintenance technician
Yearly	Replace hydraulic oil (ISO VG 46); full geometric accuracy check; replace coolant completely	Maintenance engineer

5. Chuck Mounting Bolt Torque

When re-mounting the chuck after maintenance, tighten the mounting bolts in a criss-cross sequence to **110 Nm** using a calibrated torque wrench.

6. Common Alarms

Alarm	Meaning	Operator action
AL-101	Hydraulic pressure low	Stop cycle. Check oil level and pressure gauge. Call maintenance if pressure stays below 3.5 MPa.

Alarm	Meaning	Operator action
AL-104	Way lube oil level low	Top up with ISO VG 68 oil to the MAX mark, then press RESET. Machine can continue after reset.
AL-210	Chuck clamp not confirmed	Do not start cycle. Check chuck pressure and part seating. Call maintenance if repeated.
AL-305	Spindle overload	Stop cycle. Check tool wear and cutting parameters. Inform shift supervisor.
AL-412	Coolant level low	Top up coolant mixed to 6 - 8 % concentration.

7. Replacing a Hydraulic Hose

1. Isolate the machine as per SAF-SOP-003, including releasing stored hydraulic pressure.
2. Confirm the panel pressure gauge reads zero before loosening any fitting.
3. Place a drip tray under the hose and loosen the fittings slowly.
4. Fit a new hose of the same rating and tighten fittings.
5. Remove locks as per SAF-SOP-003, start the hydraulic unit and confirm pressure returns to 3.5 - 4.0 MPa with no leaks.